ICLAB 2019 Spring Final Exam

Lecture 07(10%)

Q: Please explain how System Verilog can improve

a. Designs efficiency
b. Design/Verification communication

Q: Please explain the functions of following syntax in System Verilog and their benefits.

a. Package
b. Interface

Lecture 08(10%)

Following is a covergroup declaration.

```
covergroup final_exam @(posedge clk);
    option.per_instance = 1;
    option.at_least = 10;
    signal_0: coverpoint inf.signal_0{
        option.auto_bin_max = 32;
    }
endgroup
```

a. We know that the default value of the option per_instance is 0. What is the purpose of the option per_instance?
b. If inf.signal_0 is a 8-bit signal, how many bins will be created? And how many numbers will contain in one bin?

Lecture 09(20%)

Please **list 3 ways** that can solve clock domain crossing (CDC) problem and write down the advantage and disadvantage of them.

In Lab09, we inserted synchronizers between multi clock domains. However, without further settings, the slack will be negative in synthesis and timing violation will occur in gate level simulation. Express what action have been done to remove the negative slack issue in synthesis (5%) and timing violation issue in gate level simulation (5%).

Lecture 10(20%)

When we use clock gating in the design, which gating consumes less power? AND-gating or OR-gating? Why? Please justify your answer.

Please briefly describe 3 different voltage scaling strategies
Static Voltage Scaling(SVS)
Multi-Level Voltage Scaling(MVS)
Dynamic Voltage and Frequency Scaling(DVFS)

Lecture 11(20%)

Please explain what is cross talk, and list at least three solutions.

Please describe the benefits of using **wire group** and **interleaving** on power ring.

Lecture 12(20%)

Q1: List **at least 3 effect** toward Chip Failure as the result of IR drop.

Q2: Please briefly explain what is **Electromigration(EM)**

Extra (Chinese or English)

Q1: What is the most impressive Lab, Project, Lecture or Bonus Course in this semester, Why?

Q2: Is there any idea or proposal to improve this course?